PIC 10A 1A

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Today…

- HW2 Hints
- More Advanced/Challenging Exercises on Loops
  - Finding Matches
  - (Fibonacci/Prime Number Generator)
- (Pseudo-)Random Numbers in C++
  - Exercise: Random Walk Simulator

HW2 Hints

HW2 Hints

- Exercise 1: Consider three `if` clauses (you don't need `else`. why?)
- Exercise 2: Num2Rome → a bit more complicated
- But as we did in HW1, let's first think about a simplified problem:
  - `num` is between 1 and 9, inclusive, and we want to convert num into a Roman numeral
  - Four cases, basically:
    - `1 <= num < 4`: Print `I` `num` times
    - `num == 4`: Print `IV`.
    - `5 <= num < 9`: Print `V` first, and then `I` appropriate times (how many times?)
    - `num == 9`: Print `IX`.

  - If you can solve this simplified version, now you can extend this method.

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HW2 Hints

- Consider `num` between 1 and 99. There are two strategies:
  - A. Process ones digit first, and do `num/=10` (backward)
  - B. Process tens digit first, and do `num-=…` (forward)
- Either way is fine. For example, the pseudocode for strategy B:
  1. Ensure `num<100`
  2. Check if num has non-zero tens digit (`num>=10`?)
  3. Find the Roman numeral for this digit (using ‘X’, ‘L’, ‘C’)
  4. Subtract the corresponding number from num and check that `num<10`
  5. Similar to 2-4: check if `num>=1` and use ‘I’, ‘V’, ‘X’ to find the Roman numeral
- e.g. `num = 48`:
  - 1. and 2.: `num<100` and `num>=10`.
  - 3. and 4.: Find “XL” for 40, and `num-=40`
  - 5. `num>=1`, and find “VIII” for 8. `num-=8` and check if `num==0`

n	output
48	“”
8	“XL”
3	“XLV”
0	“XLVIII”

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HW2 Hints

- Strategy A (*backwards*): consider using the following operation:
  - `output = [something_to_add] + output;` // add a string *on the left*

- e.g. `num = 48`:
  - Take the ones digit `d = num%10`
  - Add the Roman numeral ("VIII") for `d` to the left
  - Throw ones digit away: `num/=10`
  - Compute `d = num%10` again
  - Add the Roman numeral ("XL") for `d` to the left

n	d	output
48	8	""
48	5	"V"
48	0	"VIII"
4	4	"VIII"
0	0	"XLVIII"

- This looks a bit more complicated, but in fact it's easier to "generalize"
  - See the next slides for this

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HW2 Hints: a smarter (but more difficult) method

- Challenge for Exercise 2:
- Can you solve this only using 3 if's and 2 while's?
- HINT: Consider strategy A, and the following helper variables:

```
string romans = "IVXLCDM"
int place = 0;
```

- And notice that
  - I, X, C, M → 0th, 2nd, 4th, and 6th, respectively
  - V, L, D → 1st, 3rd, and 5th, respectively
  - *i.e.*, in terms of the indices
    - 9: IX, XC, CM (02, 24, 46)
    - 4: IV, XL, CD (01, 23, 45)

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More Challenging Loop Exercises

Finding Matches

(Fibonacci/Prime Number Generator)

Finding Matches

- A code snippet finding the first match of the input character from the input string is given (see github repo)
- We will modify the program so that it finds
  - characters in case-**insensitive** manner (i.e. ‘h’ and ‘E’ are in “Hello, World!”)
  - **all** matches (i.e. prints out the position of every match)
    - ‘h’ in “Hello, World!”: 0th
    - ‘o’ in “Hello, World!”: 4th **and 8th**
    - ‘x’ in “Hello, World!”: none

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Fibonacci/Prime Number Generator

- 1. Fibonacci Number Generator
  - The user inputs a positive integer (**n**)
  - The program outputs the **n**-th Fibonacci number F_n
  - Fibonacci numbers satisfy: $F_{n+2} = F_{n+1} + F_n$
    (c.f. rabbit population puzzle posed by Fibonacci)

- 2. Prime Number Generator
  - The user inputs a positive integer (**n**)
  - The program outputs the **n**-th prime number P_n
  - First few prime numbers are: 2, 3, 5, 7, 11, 13, …

- Input and Output format:

```
Please enter a positive integer.
5
The 5th Fibonacci number is 5.
The 5th prime number is 11.
```

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Hints

- Fibonacci numbers are relatively simple
  - Just use a for loop
  - Can be done using 3 variables

- Prime numbers is more difficult
  - Break the problem into smaller pieces
  - First create a variable `int cnt` that counts the number of primes so far
  - You can then use a while loop

```
while(cnt < n) { // we want the n-th prime…
      // increment the current number
      // check if the current number is prime
      // if so, increment cnt
      // if not, don't increment cnt
```

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Hints

- Primality test (smaller piece of the entire problem)
- An integer is not prime if it has a divisor greater than 1
- So, consider using a loop to check if it has a divisor
- *Mathematical fact: you only need to check if it has a divisor $\leq \sqrt{N}$ → how to implement this w/o using `std::sqrt()`?
- You can set a flag `bool isPrime` to store the primality test result

```
for(div = 2; div*div <= num; ++div ) {
      // check if div divides num
      // if so, set isPrime false and exit from this (inner) loop
      // if not, do nothing
```

Q: what should be the *default value* for `isPrime`?

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(Pseudo-)Random Numbers in C++

rand() <cstdlib>

- `rand()` generates a (pseudo-) random integer,
  - Between `0` and `RAND_MAX`
    - RAND_MAX depends on the library
    - But at least 0x7fff = $2^{15} - 1 = 32767$ on any stand library implementation
  - Using a "seed"

- Using the same seed, `rand()` will generate the same sequence of random numbers
  - More on this later

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Using `rand()` – Floating Point numbers

- Since it generates an `int` between `0` and `RAND_MAX`, you can generate a random number between `0` and `X` (inclusive) by

  - `rand_num = (rand()*1./RAND_MAX)*X;`

    Type Conversion

    Normalized to 0 ~ 1

  - Or,

  - `rand_num = (static_cast<double>(rand())/RAND_MAX)*X;`

    which is preferable

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Using `rand()` – Floating Point numbers

- If you want a real number between A and B,
  - The size of interval = (B – A)
  - Starts from A

- `rand_num = ((rand()*1./RAND_MAX)*(B-A) + A;`

Between 0 and (B-A)

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Using `rand()` – Integers

- For integers, we can use the % (mod/remainder) operator

- Random int between 0 and N (inclusive):
  - `n = rand()%(N+1)`
  - The remainder is always between 0 and N
    - Note that, there are **N+1** different integers in [0,N]

- Random int between A and B:

  = A + (random int between 0 and (B-A))
  - `n = A + rand() % (B-A+1)`

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A Seed for `rand()`

- The function `rand()` generates a sequence of random numbers using a seed
- The seed can be set by `srand(some_number);`
- Ex) `some_number` == 1 → `rand()` gives
  - 41
  - 18467
  - 6334
  - 26500
  - 19169
  - 15724 …

  - Exercise: Use rand() several times without srand() commands, and verify that you get the sequence above

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Seed for `rand()`

- With the same seed, you'll have the same sequence of random numbers

- So, to get a random-like numbers (pseudo-random numbers), use different seed every time you run the code

- Commonly used trick: use a ***current time*** as a seed
  - `time(nullptr)` in `<ctime>` library returns the current time in **seconds**, since *00:00, Jan 1 1970 UTC*
  - Ex) `time(nullptr) == 1643894411` when this slide was created

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Summary

- #include `<cstdlib>` for `rand()` and `srand()`
- #include `<ctime>` for `time()`
- Use `time(nullptr)` to generate a seed:
  - `srand(time(nullptr));`
  - And then use `rand()`
- Intervals:
  - `r_double = (rand()*1./RAND_MAX)*(B-A) + A;`
  - `r_int = A + rand()%(B-A+1);`

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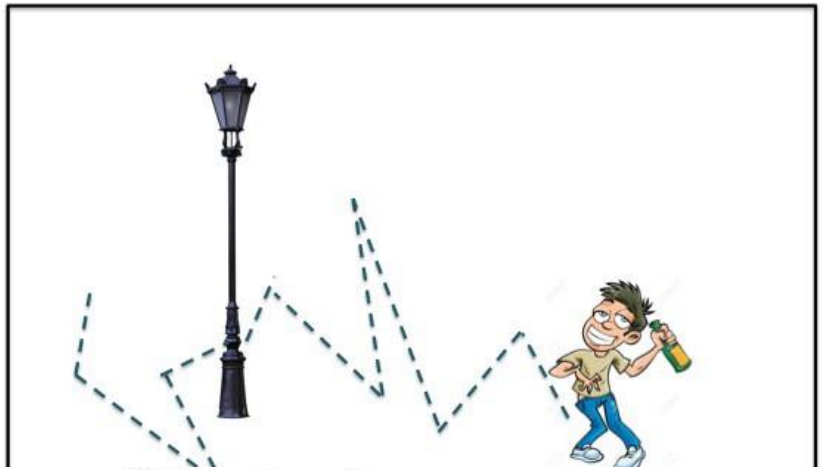
Exercise – Random Walk Simulator

(will be revisited on Thursday)

- Suppose someone walks in a constant speed, but chooses the direction randomly in each step
- The path (to be more precise, this *stochastic process*) is called a random walk



- We will simulate a 1-D random walk:
  - An object can only move forward or backward at each step
  - When the object hits the wall (upper/lower bounds), it stops walking
  - The time that it hits the wall is called the "stopping time"
  - Important in mathematical finance

Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)

Click this link

At this point, you may want to request
“more hw hints”,
“more challenging examples/exercises”,
“more details on the classroom examples”, etc.
Please let me know what you want!